Phase 19: Assumption Log

Document Code: ASM-LOG-001

Project: Duplicate Bin Ecosystem

Governance: Continuous Risk & Assumption Validation

1. Project Assumptions Ledger

Assumption ID	Category	Assumption Statement	Validation Method	Risk if Invalid	Current Status
ASM-001	Technical / Cryptography	Mathematical collision probability for SHA-256 (2^{256} space) and BLAKE2b is zero within earthly enterprise file counts.	Formal cryptographic literature & industry standard practice.	False positive deletion of non-duplicate file.	**VALIDATED**
ASM-002	Operating System	Host operating systems accurately report POSIX file modification timestamps (`st_mtime`) and file sizes (`st_size`).	Standard Python `os.stat()` platform testing across Windows, Linux, and macOS.	Persistent SQLite cache could return stale hash for modified file.	**VALIDATED**

Assumption ID	Category	Assumption Statement	Validation Method	Risk if Invalid	Current Status
ASM-003	User Behavior	Users experiencing disk space warnings prefer an automated quarantine rollback system over permanent file wiping.	Stakeholder elicitation interviews & UAT user testing feedback.	Low user adoption if customers demand instant destructive deletion without safety nets.	**VALIDATED**
ASM-004	Commercialization	Prosumer creators will purchase $4.99 Pay-As-You-Go credit packs rather than subscribing to $10/month recurring software they only use quarterly.	SaaS pricing elasticity study in docs/SAAS_ANALYSIS_AND_STRATEGY.md.	Lower lifetime value (LTV) if pay-as-you-go fails to trigger repeat purchases.	**MONITORING IN BETA**
ASM-005	Permissions	Desktop users have read permissions across their target	Pre-scan traversal error trapping (`try-except PermissionError`).	Silent scan termination if unhandled permission error halts	**VALIDATED (Handled)**

		target		error halts	

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		personal project directories.		process.	
ASM-006	Hardware Performance	Enterprise workstations possess multi-core CPUs capable of running 4 background worker threads without freezing desktop responsiveness.	Performance engine benchmark testing across dual-core and quad-core CPUs.	Desktop lag if background scanning consumes 100% CPU thread priority.	**VALIDATED**